

JOY KUMAR GHOSH

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Profile Summary

I am Computer Science graduated with a substantial research focus on **quantum information** and **secure communication**, including **quantum key distribution** and **quantum circuit simulation**. Actively involved in research writing and computational experimentation, with **two manuscripts under review**. My **Research Interests**:

- i. **Quantum Cryptography for Secure Communication**
- ii. **Quantum Machine Learning**

Education

North South University (NSU) B.Sc. in Computer Science and Engineering Distinction: magna cum laude CGPA: 3.77/4.00 ($\approx$ 94% marks)	Jan 2022 – Sep 2025 Dhaka, Bangladesh
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Birshreshtha Munshi Abdur Rouf Public College (BMAPPC) Higher Secondary Certificate (HSC) GPA: 4.33 out of 5.00 (Science)	Passing Year - 2019 Dhaka, Bangladesh
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Jadunath Pilot High School & College Secondary School Certificate (SSC) GPA: 5.00 out of 5.00 (Science)	Passing Year - 2017 Nagarpur, Tangail, Bangladesh
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Experience

Oracle APEX Web Application Developer – Intern MySoft Limited - Developed Outpatient Appointment System using APEX, and PL/SQL. [Project Link] - Participated in SRS Meetings with Client.	(3 Months) Aug 2025 – Nov 2025 Dhaka, Bangladesh
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Peer-Reviewed Journal Publications

1. “Enhanced Selection Mechanism for Combined Quantum Key Distribution Protocol.” 1st Author, under review in APL Quantum.
2. “Accelerating Noisy Quantum Circuit Simulation via Unitary Precomputation.” 1st Author, under review in AIP Advances.

Projects

1. **Undergraduate Capstone Project Showcase:** Enhanced Selection Mechanism for Entangled and Non-entangled Based Combined Quantum Key Distribution. Spring 2025. [Poster]
2. **Undergraduate CSE445 (Machine Learning) Course Project:** A thorough examination of Dhaka City’s weather data in Bangladesh, along with predictions of temperature, humidity, solar energy, solar radiation, and conditions through machine learning algorithms. Summer 2025. [Report]
3. **Undergraduate CSE465 (Pattern Recognition and Neural Network) Course Project:** Tomato Leaf Disease Detection using CNN and Lime as XAI. Spring 2025. [\[Report\]](#) [\[Repository\]](#)
4. **Undergraduate CSE425 (Concepts of Programming Language) Course Project:** Bangla Programming Language, Compiler Design with ANTLR. Summer 2024. [\[Report\]](#) [\[Repository\]](#)
5. **Undergraduate CSE311 (Database Systems) Course Project:** Nationwide Centralized Hospitals & Patients Management System, with MySQL, and PHP. Summer 2023. [\[Report\]](#) [\[Repository\]](#)

Selected Certifications

1. **Mahdy Research Academy:** Quantum Computing & Information. [Certificate]
2. **University of Colorado:** Applied Cryptography, Specialization. [[Certificate](#)]
3. **Duke University:** Explainable AI (XAI), Specialization [[Certificate](#)]
4. **KAIST:** Introduction to Quantum Information. [[Certificate](#)]

Skills

- **Quantum Tools:** Qiskit, TensorFlow Quantum.
- **Programming Language:** Python, C/C++, Java.
- **Machine Learning Tools:** PyTorch, Scikit-learn, Numpy, Pandas, LIME.
- **DevOps:** GitHub.
- **Writing Tools:** Latex and Office Suits.
- **Language:** English (Professional), Bangla (Native).

Honors & Awards

1. **Magna Cum Laude**, awarded by North South University. **Sep 2025**
2. **100% Merit-Based Scholarship** for Academic Excellence in Undergraduate. **Spring 2023 – Summer 2025**